LAB EXPERIMENT

14. Perform Perspective Transformation on the given image using python and Open CV.

Code:

```
import cv2
import numpy as np

# Read the image
img = cv2.imread(r"C:\Users\Admin\Downloads\images.jpg")

if img is None:
    print("Error: Image not found!")
else:
    # Get image dimensions
    rows, cols = img.shape[:2]

    # Define four points in the original image
    pts1 = np.float32([
        [50, 50],
        [cols - 50, 50],
        [50, rows - 50],
        [cols - 50, rows - 50]
    ])

    # Define four points in the output image
    pts2 = np.float32([
        [0, 0],
        [cols, 0],
        [0, rows],
        [cols, rows]
    ])

    # Calculate perspective transformation matrix
    matrix = cv2.getPerspectiveTransform(pts1, pts2)

    # Apply perspective transformation
    transformed = cv2.warpPerspective(img, matrix, (cols, rows))
```

```
# Display images
cv2.imshow("Original Image", img)
cv2.imshow("Perspective Transformed Image", transformed)

cv2.waitKey(0)
cv2.destroyAllWindows()
```



